



Deploying Cloud Native Qumulo (CNQ) on AWS with Terraform release v7.5

This repository contains Terraform which deploys S3 buckets and a CNQ cluster with 3 or more instances that adhere to the [AWS Well-Architected Framework](#). NOTE: as of version 7.0 this Terraform uses the Qumulo AWS Provider. This greatly simplifies Terraform operations and makes compute elasticity changes even more elegant. To learn more about the provider read the [provider docs](#).

Configure Terraform backend.tf

This Terraform defaults to store state on S3. As such the `./backend.tf` must both be configured with the S3 bucket and S3 bucket region you choose to store state in. Note you can't put variables in the backend.tf config, which is why this note is here. If you call this Terraform as a module the backend.tf will be ignored.

Getting Started with Cloud Native Qumulo (CNQ)

For an overview of CNQ, its reference architecture, and limits, see [How Cloud Native Qumulo Works](#) and for prerequisites and detailed instructions, see [Deploying Cloud Native Qumulo on AWS with Terraform](#) on the Documentation Portal.

Qumulo Core $\geq$ 7.9.2.1 is required for this Terraform

Tip: For help with deployment, configuration, updates, scaling out your cluster, and best practices for high performance, [message us on Slack](#).

Standard Deployment Example

```
module "cloud_native_qumulo" {
  source = "git::https://github.com/Qumulo/qumulo-terraform-aws.git?ref=v7.6"
  # ***** QUMULO PROVIDER VARIABLES *****
  #-----REQUIRED-----
  deployment_name = "my-deployment-name"
  ec2_key_pair    = "my-key-pair"
  instance_type   = "i7i.2xlarge"
  region          = "us-west-2"
  subnet_ids      = ["subnet-0123456789abcdef1"]
  vpc_id          = "vpc-0123456789abcdef1"

  #-----OPTIONAL-----
  additional_security_group_ids = null
  allow_cidrs                   = null
  ami_id                        = null
  cluster_iam_role_arn          = null
  cluster_security_group_id     = null
  kms_key_id                    = null
  permissions_boundary_arn      = null
  provisioner_ami_id            = null
  provisioner_iam_role_arn      = null
  provisioner_instance_type     = null
  provisioner_security_group_id = null
  s3_gateway_validation         = null
  s3_log_bucket_name            = null
  s3_log_bucket_prefix          = null
  tags = {
    owner       = "owner"
    department  = "department"
    purpose     = "purpose"
    long_running = "true"
  }

  # ***** Qumulo Cluster Variables *****
  #-----REQUIRED-----
```

```
admin_pwd_or_secrets_arn = "arn:aws:secretsmanager:us-west-2:<aws account
number>:secret:/<secret path>/<secret_name>"
cluster_name              = "CNQ-HOT"
cluster_product_type      = "HOT"
deletion_protection       = true
node_count                = 3

#-----OPTIONAL-----
audit_logging              = false
cluster_version            = null
floating_ip_count          = 12
ip_v4_or_v6               = "v4"
nexus_registration_key     = null
provider_timeout_minutes  = 30
soft_capacity_limit_tb     = null
storage_class              = null
}

output "outputs_cloud_native_qumulo" {
  value = module.cloud_native_qumulo
}
```

Terraform Documentation

 **Note:** This repository uses documentation generated with Terraform-Docs.

Requirements

Name	Version
terraform	>= 1.11.0
aws	>= 5.32
null	>= 3.1
qumulo	>= 1.4.10
random	>= 3.1

Inputs

Name	Description	Type	Default	Required
additional_security_group_ids	OPTIONAL: Use for adding custom ingress rules beyond those the provider creates	<code>list(string)</code>	<code>null</code>	no
admin_pwd_or_secrets_arn	Provide either a plaintext administrator password or an AWS Secrets Manager ARN.	<code>string</code>	n/a	yes
allow_cidrs	OPTIONAL: CIDR blocks allowed to access the cluster	<code>list(string)</code>	<code>null</code>	no
ami_id	OPTIONAL: AMI ID for cluster nodes. If omitted, the default Qumulo AMI (Ubuntu 24.04) is used. Supports Ubuntu and RHEL 8, 9, and 10 AMIs. See aws-custom-images.md for details.	<code>string</code>	<code>null</code>	no
audit_logging	OPTIONAL: Send Qumulo audit logs to AWS CloudWatch	<code>bool</code>	<code>false</code>	no

Name	Description	Type	Default	Required
	logs.			
cluster_dns_name	OPTIONAL: DNS Name for the cluster. Leave null to automatically pickup the QDNS or NLB DNS name.	string	null	no
cluster_fqdn	OPTIONAL: Fully qualified domain name for Qumulo DNS	string	null	no
cluster_iam_role_arn	OPTIONAL: When set, the provider performs no IAM writes for that role — no creation, policy updates, tagging, or deletion — and instead launches instances with your role's instance profile	string	null	no
cluster_name	Cluster name (2-15 alphanumeric characters). Used as a prefix for AWS resources created for this cluster.	string	n/a	yes
cluster_product_type	Cluster storage product type (immutable after creation). HOT: Optimized for frequently accessed data. COLD: Optimized for archival/infrequently accessed data.	string	n/a	yes
cluster_security_group_id	OPTIONAL: Bring-your-own security group ID for cluster nodes. When set, the provider will not create or modify a cluster security group; you must pre-create it in the VPC owner account with the rules documented in the Bring-your-own Security Groups guide. Required together with <code>provisioner_security_group_id</code> . Required for RAM-shared subnets where the consumer account cannot create security groups in the owner's VPC.	string	null	no
cluster_version	OPTIONAL: Qumulo software version. Defaults to latest. Immutable after creation. Upgrade version via cluster UI/API.	string	null	no
deletion_protection	Enables EC2 Termination protection and prevents Terraform from destroying non-empty S3 Buckets	bool	true	no

Name	Description	Type	Default	Required
deployment_name	Cluster name (2-15 alphanumeric characters). Used as a prefix for AWS resources created for this cluster.	string	n/a	yes
ec2_key_pair	EC2 key pair name for SSH access to cluster nodes	string	n/a	yes
floating_ip_count	OPTIONAL: Number of floating IPs. Must be 0, or between 3 and 100. NOT applicable with multi-AZ deployments.	number	3	no
instance_type	EC2 instance type for the cluster. Prefer i7i, i4i, i7ien. Supported families include m6idn, m6i, m7i, i3en, i4i, i7i, i7ie.	string	n/a	yes
ip_v4_or_v6	OPTIONAL: Only change this from default (v4) if connecting to the cluster via IPv6. Not supported for clusters deployed without the Qumulo Terraform Provider.	string	"v4"	no
kms_key_id	OPTIONAL: KMS key ARN for encrypting AWS services like EBS and S3 (immutable after creation). Qumulo encrypts all data independent of AWS default or KMS keys.	string	null	no
networking_mode	OPTIONAL: Only change this from default (host_managed) if importing a cluster that was deployed without the Qumulo Terraform Provider. Terraform versions prior to 7.0.	string	"host_managed"	no
nexus_api_token	OPTIONAL: Qumulo Nexus API token for NeuralProtect	string	null	no
nexus_registration_key	OPTIONAL: Qumulo Nexus registration key for remote support	string	null	no
nlb_cross_zone	OPTIONAL: AWS NLB Enable cross-AZ load balancing	bool	false	no
nlb_deletion_protection	Enables NLB Termination protection and prevents Terraform from destroying the NLB	bool	true	no
nlb_dns_client_affinity	OPTIONAL: AWS NLB DNS Client Routing Policy Zonal Affinity	string	"availability_zone_affinity"	no
nlb_override_subnet_ids	OPTIONAL: Private Subnet ID(s) for NLB if deploying in	list(string)	null	no

Name	Description	Type	Default	Required
	subnet(s) other than subnet(s) the cluster is deployed in			
nlb_provision	OPTIONAL: Provision an AWS NLB in front of the Qumulo cluster for load balancing and client failover	bool	false	no
nlb_public	OPTIONAL: Makes the NLB for the cluster public, setting this to true will allow anyone to reach the cluster. Not recommended for production clusters.	bool	false	no
nlb_stickiness	OPTIONAL: AWS NLB sticky sessions	bool	true	no
node_count	Number of nodes in the cluster. Valid values: 1 (single node), or 3-24. 1 and 4 not allowed for multi-AZ.	number	n/a	yes
node_hooks_files	OPTIONAL: Pre and post userdata hooks files for Cluster Nodes.	map(string)	null	no
np_deletion_protection	OPTIONAL: Causes Terraform to throw an error upon destroy for the NeuralProtect resource. Safeguard your NeuralProtect instance. Set to false to destroy.	bool	true	no
np_instance_type	OPTIONAL: NeuralProtect EC2 instance type.	string	"m6a.xlarge"	no
np_provision	OPTIONAL: true/false to enable deployment of NeuralProtect.	bool	false	no
permissions_boundary_arn	OPTIONAL: IAM permissions boundary ARN applied to cluster and provisioner roles	string	null	no
provider_timeout_minutes	The total time after which Terraform will abandon the provider deployment of the Qumulo cluster and timeout. In minutes.	number	30	no
provisioner_ami_id	OPTIONAL: AMI ID for the provisioner instance. Defaults to Ubuntu 24.04 AMI.	string	null	no
provisioner_hooks_files	OPTIONAL: Pre and post userdata hooks files for the provisioner.	map(string)	null	no
provisioner_iam_role_arn	OPTIONAL: When set, the provider performs no IAM writes for that role — no creation, policy updates, tagging, or deletion — and	string	null	no

Name	Description	Type	Default	Required
	instead launches instances with your role's instance profile			
<code>provisioner_instance_type</code>	OPTIONAL: EC2 instance type for the provisioner VM (used during deploy operations). Default: m5.xlarge.	<code>string</code>	<code>"m5.xlarge"</code>	no
<code>provisioner_security_group_id</code>	OPTIONAL: Bring-your-own security group ID for the provisioner instance. Required together with <code>cluster_security_group_id</code> . See <code>cluster_security_group_id</code> .	<code>string</code>	<code>null</code>	no
<code>r53_second_subnet_id</code>	OPTIONAL: A second subnet ID, in a unique AZ other than the cluster AZ, for single AZ clusters. This is then used to build a R53 Resolver to forward traffic to Qumulo DNS for Floating IP resolution.	<code>string</code>	<code>null</code>	no
<code>region</code>	AWS region for deployment	<code>string</code>	<code>n/a</code>	yes
<code>s3_gateway_validation</code>	OPTIONAL: Terraform will test for an S3 gateway if this is left at null or set to true. Set to false for environments that have complex VPC networking.	<code>bool</code>	<code>null</code>	no
<code>s3_log_bucket_name</code>	OPTIONAL: Bucket name for S3 logging	<code>string</code>	<code>null</code>	no
<code>s3_log_bucket_prefix</code>	OPTIONAL: Bucket prefix for S3 logging	<code>string</code>	<code>null</code>	no
<code>soft_capacity_limit_tb</code>	OPTIONAL: Soft capacity limit in TB (50 to 50000). Default is 500TB. Can be increased to add storage, but cannot be decreased. It's like a quota, unused capacity is not billed.	<code>number</code>	<code>500</code>	no
<code>storage_class</code>	HOT cluster default is INTELLIGENT_TIERING or STANDARD, COLD cluster default is GLACIER_IR or STANDARD_IA	<code>string</code>	<code>null</code>	no
<code>subnet_ids</code>	Subnet ID for a single AZ deployment or 3+ subnets, one per AZ, for a multi-AZ deployment)	<code>list(string)</code>	<code>n/a</code>	yes
<code>tags</code>	OPTIONAL: Tags to apply to all AWS resources created for this cluster.	<code>map(string)</code>	<code>null</code>	no
<code>vpc_id</code>	VPC ID for the cluster	<code>string</code>	<code>n/a</code>	yes

Outputs

Name	Description
cluster_name	Name of the Qumulo cluster
cluster_soft_capacity_limit_tb	Total capacity the cluster may consume. Only used capacity is billed.
cluster_uuid	UUID of the Qumulo cluster
deployment_unique_name	Unique deployment identifier
endpoint_ips	Client-facing IPs. Floating IPs if configured, otherwise primary IPs.
endpoints	Connection endpoints for various protocols
primary_ips	Per-node primary IPs. Use these directly when no floating IPs are configured, or for per-node access.
provisioner_log	CloudWatch Log for the provisioner

About This Repository

This repository uses the [MIT license](#). All contents Copyright © 2024 [Qumulo, Inc.](#), except where specified. All trademarks are property of their respective owners.

For more information about this repository, contact [Dack Busch](#) and [Gokul Kupparaj](#).